Sets and functions

COMP0147

Elaine Pimentel & Mário Sérgio Alvim

Term 1 – 2026/27



Sets: Introduction

- **Sets** are the fundamental discrete structures on which all other discrete structures can be built.

- **Set Theory** is capable of representing all of Mathematics.

  Basic concepts such as sets, membership of elements in a set, the empty set, and operations on sets (union, intersection, complement, ...) can capture concepts such as arithmetic, logic, etc.

- The concepts we study here are essential to several areas, including some that we will study in this course:

  - functions,
  - sequences,
  - sums and products,
  - combinatorics,
  - relations.
  - ...

Sets

Sets

- A **set** is an unordered collection of well-defined objects, called the **elements** or **members** of the set.

  We write

  $$a \in A$$

  to denote that the element a belongs to the set A.

  We write

  $$a \notin A$$

  to denote that the element a does not belong to the set A.

- We usually use uppercase letters to denote sets, and lowercase letters to denote elements of these sets.

Ways to define a set

- Listing its elements inside **curly braces**:

  $S_1 = \{\texttt{Ana}, \{\texttt{Bia}\}, \texttt{Carlos}\}$ $\quad$ $S_2 = \{\texttt{Jupiter}, \pi, \texttt{Ana}, \{200\}\}$

- Specifying a **property** that defines its elements, as in $S = \{x \mid P(x)\}$:

  $S_3 = \{x \in \mathbb{R} \mid -2 \leq x \leq 5\}$ $\quad$ $S_4 = \{x \in \mathbb{N} \mid x \text{ is prime and } x > 431\}$

- Using a **recursive definition**:

  $$\begin{cases} 1 \in S_5, \\ \text{if } x \in S_5 \text{ and } x + 2 < 10, \text{ then } x + 2 \in S_5. \end{cases}$$

Ways to define a set

- Defining a **characteristic function**:

$$\mu_A(x) = \begin{cases} 1, & \text{if } x \in \mathbb{N} \text{ and } x \text{ is prime,} \\ 0, & \text{otherwise.} \end{cases}$$

- It is not always possible to use every kind of definition:

  1. It is not possible to define the set $S = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$ by listing all its elements.

Some important sets

- Some important sets are:

1. $\mathbb{N} = \{0, 1, 2, 3, 4, 5, ...\}$ is the set of **natural numbers**.
2. $\mathbb{Z} = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$ is the set of **integers**.
3. $\mathbb{Z}^+ = \{1, 2, 3, 4, 5 ...\}$ is the set of **positive integers**.
4. $\mathbb{Q} = \{p/q \mid p \in \mathbb{Z}, q \in \mathbb{Z}, \text{ and } q \neq 0\}$ is the set of **rational numbers**.
5. $\mathbb{R}$ is the set of **real numbers**.
6. $\mathbb{R}^+$ is the set of **positive real numbers**.
7. $\mathbb{C}$ is the set of **complex numbers**.

Set equality

- Two sets are **equal** iff they have the same elements.

  Formally, for all sets A and B,

$$A = B \quad \leftrightarrow \quad \forall x : (x \in A \ \leftrightarrow \ x \in B).$$

- The definition of set equality implies that:
  - The order in which the elements are listed is irrelevant:
    1. $\{a, b, c, d\} = \{d, c, b, a\} = \{c, a, d, b\}$
  - Elements may appear more than once in a set:
    1. $\{a, a, a, a, b, b, b, c, c, d\} = \{a, b, c, d\}$

Subsets

- A set A is called a **subset** of a set B iff every element of A is also an element of B.

  We use $A \subseteq B$ to denote that A is a subset of B.

- Formally:

$$A \subseteq B \quad \leftrightarrow \quad \forall x : (x \in A \ \rightarrow \ x \in B).$$

- The phrases "A is **contained in** B" and "B **contains** A" are alternative ways of saying that A is a subset of B.

1. The set of natural numbers is a subset of the integers.

2. The set of Brazilians is a subset of the set of Brazilians. (There is nothing to prevent a set from being a subset of itself!)

3. The set of complex numbers is not a subset of the real numbers.

Proper subsets

- A set A is a **proper subset** of a set B iff every element of A is in B and there is at least one element of B that is not in A.
- Formally:

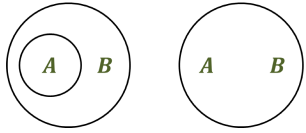
$$\begin{aligned} A \subset B \quad &\leftrightarrow \quad \forall x : (x \in A \ \rightarrow \ x \in B) \quad \wedge \quad \exists x : (x \in B \ \wedge \ x \notin A) \\ &\leftrightarrow \quad A \subseteq B \quad \wedge \quad A \neq B. \end{aligned}$$

1. The set of natural numbers is a proper subset of the set of integers.
2. The set of Brazilians is not a proper subset of the set of Brazilians.

Venn diagrams

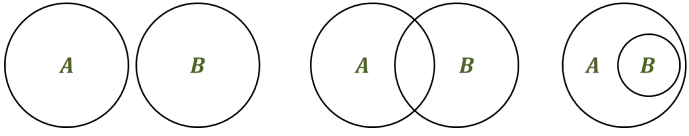
- If the sets A and B are represented by regions in the plane, relations between A and B can be represented by drawings called **Venn diagrams**.

- Example 1: $A \subseteq B$.



□

- Example 2: $A \nsubseteq B$.



□

The empty set

- The **empty set** or **null set** contains no elements.

  We denote the empty set by $\{\}$ or $\emptyset$.

- Note that $\{\emptyset\}$ does not denote the empty set, but rather the set whose only element is the empty set.

- **Theorem**: The empty set is a subset of every set.

  Proof. Let A be any set. Then

$$\forall x : (x \in \emptyset \ \rightarrow \ x \in A)$$

is true vacuously, since the premise of the implication is always false.

Hence $\emptyset \subseteq A$. $\square$

Power set

- Given a set A, the **power set of** A is the set of all subsets of A.

  We denote the power set of A by $\mathscr{P}(A)$.

- Examples:

  1. Given the set $S = \{x, y, z\}$, its power set is

Power set

- Given a set A, the **power set of** A is the set of all subsets of A.

  We denote the power set of A by $\mathscr{P}(A)$.

- Examples:

  1. Given the set $S = \{x, y, z\}$, its power set is

$$\mathscr{P}(S) = \{\emptyset, \{x\}, \{y\}, \{z\}, \{x, y\}, \{x, z\}, \{y, z\}, \{x, y, z\}\}.$$

2. Given the empty set $\emptyset$, its power set is

$$\mathscr{P}(\emptyset) = \{\emptyset\}.$$

Power set

- **Theorem:** If a finite set A has n elements, then $\mathscr{P}(A)$ has 2^n elements.

  Proof. To form any subset S of A, we can consider each element $a_i \in A$ $(1 \leq i \leq n)$, deciding whether $a_i \in S$ or $a_i \notin S$.

  Since there are two choices for each element (it belongs or does not belong), and there are n elements in A, there are 2^n ways to form a subset S of A.

  Hence, $|\mathscr{P}(A)| = 2^n$. □

Ordered tuples

- An **ordered n-tuple** $(a_1, a_2, \ldots, a_n)$ is an ordered collection of n elements, where a_1 is the first element, a_2 is the second element, ..., and a_n is the nth element.
- Some ordered n-tuples have special names:
  - An ordered 2-tuple is called an **ordered pair**.
  - An ordered 3-tuple is called an **ordered triple**.
  - An ordered 4-tuple is called an **ordered quadruple**.
  - ...
- Two ordered n-tuples $(x_1, x_2, \ldots, x_n)$ and $(y_1, y_2, \ldots, y_n)$ are **equal** iff

$$x_i = y_i, \quad \text{for } i = 1, \ldots n.$$

Cartesian product

- Let A and B be sets. The **Cartesian product** of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$.

  Formally:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

- Example 3: Let $A = \{1, 2\}$ and $B = \{a, b, c\}$.

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

$$B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$$

$$A \times A = A^2 = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$$

Note that, in general, $A \times B \neq B \times A$. □

Cartesian product

- Cartesian products can be generalised to more than two sets.
- Let $A_1, A_2, \ldots, A_n$ be sets.

  The **Cartesian product** of $A_1, A_2, \ldots, A_n$, denoted by

$$A_1 \times A_2 \times \ldots \times A_n,$$

is the set of all ordered n-tuples $(a_1, a_2, \ldots, a_n)$, where $a_i \in A_i$ for $i = 1, \ldots, n$.

Formally:

$$A_1 \times A_2 \times \ldots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A_i \text{ for } i = 1, \ldots, n\}$$

- Example 4: Let $A = \{0, 1\}$, $B = \{a, b\}$, and $C = \{\gamma, \delta\}$.

$$\begin{aligned} A \times B \times C \;=\; & \{(0, a, \gamma), (0, a, \delta), (0, b, \gamma), (0, b, \delta), \\ & (1, a, \gamma), (1, a, \delta), (1, b, \gamma), (1, b, \delta)\} \end{aligned}$$

□

The size of finite sets

- Let A be a finite set containing exactly n distinct elements.

  We say that the **cardinality** (or **size**) of A is n.

  The notation $|A| = n$ indicates that the size of A is n elements.

- We will study the cardinality of infinite sets later.

Operations on sets

Operations on sets: Introduction

- Two or more sets can be combined in different ways.

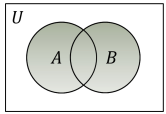
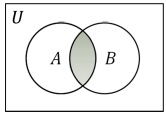
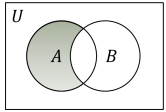
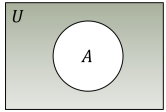
  For example, given the set of Discrete Mathematics students and the set of people born in Minas Gerais, we can define:

  1. the set of people from Minas Gerais who study Discrete Mathematics,
  2. the set of people who are from Minas Gerais or study Discrete Mathematics,
  3. the set of Discrete Mathematics students who are not from Minas Gerais,
  4. ...

- Here we study **operations on sets** that allow us to create more complex sets from simpler ones.

Operations on sets

- Let A and B be subsets of the universal set U:

Union:	$A \cup B = \{x \in U \mid x \in A \ \vee \ x \in B\}$	
Alternatively:	$x \in A \cup B \leftrightarrow x \in A \ \vee \ x \in B$	
Notation:	$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \ldots \cup A_n$	
Intersection:	$A \cap B = \{x \in U \mid x \in A \ \wedge \ x \in B\}$	
Alternatively:	$x \in A \cap B \leftrightarrow x \in A \ \wedge \ x \in B$	
Notation:	$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \ldots \cap A_n$	
Difference:	$A - B = \{x \in U \mid x \in A \ \wedge \ x \notin B\}$	
Alternatively:	$x \in A - B \leftrightarrow x \in A \ \wedge \ x \notin B$	
Complement:	$\overline{A} = \{x \in U \mid x \notin A\}$	
Alternatively:	$x \in \overline{A} \leftrightarrow x \notin A$	

Operations on sets

Example 5: Let $A = \{1, 3, 4, 5\}$ and $B = \{1, 2, 5, 6\}$.

Take the universal set to be $U = \{0, 1, 2, 3, 4, 5, 6, 7\}$.

- $A \cup B = \{1, 2, 3, 4, 5, 6\}$
- $A \cap B = \{1, 5\}$
- $A - B = \{3, 4\}$
- $B - A = \{2, 6\}$
- $\overline{A} = \{0, 2, 6, 7\}$
- $\overline{B} = \{0, 3, 4, 7\}$

□

Set equality

- Two sets A and B are equal if and only if every element of A is in B, and every element of B is in A.

- A convenient way to show that two sets are equal is to show that each set is a subset of the other.

  Formally:

$$A = B \quad \text{iff} \quad \forall x : (x \in A \ \leftrightarrow \ x \in B).$$

Theorem: $A = B$ iff $A \subseteq B$ and $B \subseteq A$.

Proof. Writing $A \subseteq B$ and $B \subseteq A$ formally:

$$\begin{aligned}
& A = B \\
\equiv \ & \forall x : (x \in A \leftrightarrow x \in B) && \text{(definition of equality)} \\
\equiv \ & \forall x : ((x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)) && \text{(definition of } \leftrightarrow\text{)} \\
\equiv \ & (\forall x : (x \in A \rightarrow x \in B)) \wedge (\forall x : (x \in B \rightarrow x \in A)) && \text{(distributivity of } \forall \text{ over } \wedge\text{)} \\
\equiv \ & A \subseteq B \wedge B \subseteq A && \text{(definition of } \subseteq\text{)}
\end{aligned}$$

□

Set equality

Example 6: Show that $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$.

Solution.

- Method 1: Manipulating the definitions of the set operators.

  We will show that $x \in \overline{(A \cap B)}$ iff $x \in (\overline{A} \cup \overline{B})$:

$$
\begin{array}{lcll}
x \in \overline{(A \cap B)} & \leftrightarrow & x \notin (A \cap B) & \text{(definition of complement)} \\
& \leftrightarrow & \neg(x \in (A \cap B)) & \text{(definition of } \notin\text{)} \\
& \leftrightarrow & \neg((x \in A) \wedge (x \in \cap B)) & \text{(definition of intersection)} \\
& \leftrightarrow & \neg(x \in A) \vee \neg(x \in B) & \text{(De Morgan's law)} \\
& \leftrightarrow & (x \notin A) \vee (x \notin B) & \text{(definition of } \notin\text{)} \\
& \leftrightarrow & (x \in \overline{A}) \vee (x \in \overline{B}) & \text{(definition of complement)} \\
& \leftrightarrow & x \in (\overline{A} \cup \overline{B}) & \text{(definition of union)}
\end{array}
$$

Set equality

Example 6: (Continuation)

- Method 2: Using a **membership table**, where we use the symbol 1 to indicate that an element belongs to a set, and the symbol 0 to indicate that it does not.

  The table below shows that an element belongs to $\overline{(A \cap B)}$ (the fourth column) iff it belongs to $\overline{A} \cup \overline{B}$ (the seventh column):

A	B	$A \cap B$	$\overline{(A \cap B)}$	$\overline{A}$	$\overline{B}$	$\overline{A} \cup \overline{B}$
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

□

Set equality

- Let all the sets below be subsets of the universal set U.

Commutativity	$A \cap B = B \cap A$	$A \cup B = B \cup A$
Associativity	$(A \cap B) \cap C = A \cap (B \cap C)$	$(A \cup B) \cup C = A \cup (B \cup C)$
Distributivity	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Un. and int. with U	$A \cap U = A$	$A \cup U = U$
Double complement	$\overline{\overline{A}} = A$	
Idempotence	$A \cap A = A$	$A \cup A = A$
De Morgan	$\overline{(A \cap B)} = \overline{A} \cup \overline{B}$	$\overline{(A \cup B)} = \overline{A} \cap \overline{B}$
Absorption	$A \cap (A \cup B) = A$	$A \cup (A \cap B) = A$
Set difference	$A - B = A \cap \overline{B}$	
Un. and int. with $\varnothing$	$A \cup \varnothing = A$	$A \cap \varnothing = \varnothing$
Un. and int. with the comp.	$A \cup \overline{A} = U$	$A \cap \overline{A} = \varnothing$
Complements of U and $\varnothing$	$\overline{U} = \varnothing$	$\overline{\varnothing} = U$

Disjoint sets

- Two sets are called **disjoint** iff they have no elements in common.

  Formally:

  $$A \text{ and } B \text{ are disjoint} \quad \leftrightarrow \quad A \cap B = \emptyset.$$

- **Proposition:** Given two sets A and B, $(A - B)$ and B are disjoint.

  Proof. By contradiction. Suppose the statement is false; that is, suppose there are sets A and B such that $(A - B) \cap B \neq \emptyset$. Then there is an element x such that $x \in (A - B) \wedge x \in B$. In particular, this means that $x \in B$.

  On the other hand, we also have $x \in (A - B)$, which, by the definition of difference, means that $x \in A \wedge x \notin B$. In particular, this implies that $x \notin B$.

  Thus we reach a contradiction, since $x \in B$ and $x \notin B$. Therefore, the proposition must be true. $\square$

Partitions of a set

- The sets $A_1, A_2, \ldots, A_n$ are called **mutually disjoint** (or **pairwise disjoint**, or **non-overlapping**) iff $A_i \cap A_j = \emptyset$ for all $i, j = 1, 2, \ldots, n$ with $i \neq j$.
- A collection of nonempty sets $\{A_1, A_2, \ldots, A_n\}$ is a **partition** of the set A iff
  1. $A = A_1 \cup A_2 \cup \ldots \cup A_n$, and
  2. $A_1, A_2, \ldots, A_n$ are mutually disjoint.
- Example 7: $\{\{1, 2, 5\}, \{3\}, \{4\}\}$ is a partition of $\{1, 2, 3, 4, 5\}$.

□

- Example 8: $\mathbb{N}$ can be partitioned into the set of even numbers and the set of odd numbers.

□

Russell's Paradox and "Naive" Set Theory

The Barber Paradox

- We have seen that a set can be specified by a property that defines it, as in $S = \{x \mid P(x)\}$:

  1. $\{x \in \mathbb{R} \mid -2 \leq x \leq 5\}$
  2. $\{x \in \mathbb{N} \mid x \text{ is prime and } x > 431\}$

- However, the property $P(x)$ cannot be an arbitrary property.

- **Barber Paradox:** "The barber is someone who shaves all and only those men who do not shave themselves."

  The question is: does the barber shave himself?

  Equivalently, let b be the barber and let B be the set of all people whom barber b shaves. Then:

  $$b \in B?$$

The Barber Paradox

- We have seen that a set can be specified by a property that defines it, as in $S = \{x \mid P(x)\}$:

  1. $\{x \in \mathbb{R} \mid -2 \leq x \leq 5\}$
  2. $\{x \in \mathbb{N} \mid x \text{ is prime and } x > 431\}$

- However, the property $P(x)$ cannot be an arbitrary property.

- **Barber Paradox:** "The barber is someone who shaves all and only those men who do not shave themselves."

  The question is: does the barber shave himself?

  Equivalently, let b be the barber and let B be the set of all people whom barber b shaves. Then:

  $$b \in B?$$

Paradox: $b \in B \leftrightarrow b \notin B$!

Russell's Paradox and "Naive" Set Theory

- The Barber Paradox is a special case of the problem identified by Bertrand Russell:

  Russell's Paradox: If we define S as "the set of all sets that do not contain themselves as an element", that is,

  $$S = \{A \mid A \text{ is a set and } A \notin A\},$$

  how do we decide whether $S \in S$?

  Paradox: $S \in S \leftrightarrow S \notin S$!

- Lessons:
  - "Naive" Set Theory ("naive set theory") cannot be used carelessly.
  - For this reason, there are **axiomatic approaches**, such as Zermelo–Fraenkel set theory ("ZF set theory").

Functions

Functions: Introduction

- We often need to assign a particular element of one set to each element of another set. For example, we may:

  1. Assign a grade of A, B, C, D, E or F to each Discrete Mathematics student.
  2. Assign to each integer its square.
  3. Assign to each country its head of state.

- The concept of a **function** formalises this kind of assignment.
- In mathematics and computer science, functions are fundamental:
  - in defining discrete structures such as **sequences** and **strings**,
  - in studying the complexity of algorithms,
  - in constructing recursive algorithms,

Functions

- Let A and B be non-empty sets. A **function** f from A to B assigns exactly one element of B to each element of A. We write

$$f(a) = b$$

if b is the unique element of B assigned by f to the element a of A.

Functions

- If f is a function from A to B, we write

$$f : A \to B$$

to denote the **type** of the function. The set A is called the **domain** of f. The set B is called the **codomain** of f. The **image** (or **range**) of f is the set of values that f can take:

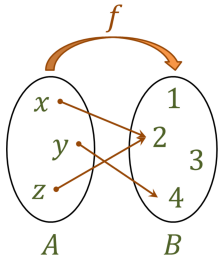
$$\text{image of } f \ = \{b \in B \mid b = f(a) \text{ for some } a \in A\}$$

The **preimage** of an element $b \in B$ is the set of elements $a \in A$ mapped to b by f:

$$\text{preimage of } b \ = \{a \in A \mid f(a) = b\}$$

Functions: Examples

Example 9: Let $A = \{x, y, z\}$ and $B = \{1, 2, 3, 4\}$. Let the function $f : A \to B$ be defined by the diagram below.



- Domain of f: $\{x, y, z\}$; Codomain of f: $\{1, 2, 3, 4\}$; Image of f: $\{2, 4\}$
- $f(x) = 2, f(y) = 4; f(z) = 2$
- Preimage of 1: $\emptyset$; Preimage of 2: $\{x, z\}$; Preimage of 3: $\emptyset$; Preimage of 4: $\{y\}$
- The function f can be represented as a **set of ordered pairs**:

$$f = \{(x, 2), (y, 4), (z, 2)\}$$

□

Functions: Examples

- Example 10: Further examples of functions:

  - The square function $f : \mathbb{R} \to \mathbb{R}$:

$$f(x) = x^2 \quad \text{or} \quad f : x \mapsto x^2$$

- The successor function $f : \mathbb{Z} \to \mathbb{Z}$:

$$f(n) = n + 1 \quad \text{or} \quad f : n \mapsto n + 1$$

- A constant function $f : \mathbb{Q} \to \mathbb{Z}$:

$$f(r) = 2 \quad \text{or} \quad f : r \mapsto 2$$

- The father function $f : P \to P$, where P is the set of all people:

$$f : p \mapsto p', \text{ where } p' \text{ is the father of } p$$

Equality of functions

- Two functions f and g are **equal** if and only if they:
  - have the same domain,
  - have the same codomain,
  - map each element of the domain to the same element of the codomain.

  Formally, for two functions $f, g : A \to B$:

$$f = g \qquad \text{iff} \qquad \forall a \in A : f(a) = g(a).$$

- Example 11: Let $f(x) = |x|$ and $g(x) = \sqrt{x^2}$. Then $f = g$, since for every $x \in \mathbb{R}$:

$$|x| \;=\; \sqrt{x^2}.$$

□

Injective functions

- A function $f : A \to B$ is **injective** (or **one-to-one**) if and only if, for all $a_1, a_2 \in A$:

$$a_1 \neq a_2 \quad \to \quad f(a_1) \neq f(a_2)$$

or, equivalently,

$$f(a_1) = f(a_2) \quad \to \quad a_1 = a_2$$

- Intuitively, a function is injective if distinct elements of the domain are mapped to distinct elements of the codomain.

- Example 12: In each case below, let $f : \mathbb{R} \to \mathbb{R}$. Then:
  - $f(x) = x + 1$ is injective;
  - $f(x) = x/10$ is injective;
  - $f(x) = x^2$ is not injective;

□

Surjective functions

- A function $f : A \to B$ is **surjective** (or **onto**) if and only if, for every $b \in B$, there exists an $a \in A$ such that $f(a) = b$.
- Intuitively, a function is surjective if each element of the codomain is the image of at least one element of the domain.
- Example 13: In each case below, let $f : \mathbb{R} \to \mathbb{R}$. Then:
  - $f(x) = x + 1$ is surjective;
  - $f(x) = x/10$ is surjective;
  - $f(x) = x^2$ is not surjective;

□

Bijective functions

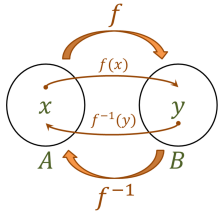
- A function $f : A \to B$ is **bijective** if and only if it is both injective and surjective.
- Example 14: In each case below, let $f : \mathbb{R} \to \mathbb{R}$. Then:
  - $f(x) = x + 1$ is bijective, since it is injective and surjective;
  - $f(x) = x/10$ is bijective, since it is injective and surjective;
  - $f(x) = 2^x$ is not bijective (it is injective but not surjective);
  - $f(x) = (x-1)(x-2)(x-3)$ is not bijective (it is surjective but not injective);

□

Inverse functions

- Let $f : A \to B$ be a bijective function. The **inverse function** of f is the function $f^{-1} : B \to A$ such that

$$f^{-1}(y) = x \quad \text{iff} \quad y = f(x)$$



- Example 15: The function $f : \mathbb{Z} \to \mathbb{Z}$ defined by $f(x) = x + 1$ is invertible because it is bijective. Its inverse is

$$f^{-1}(x) = x - 1$$

□

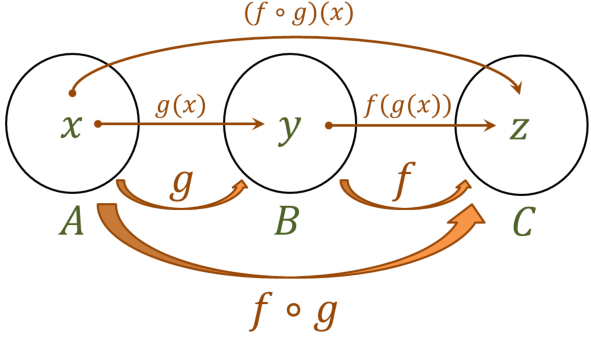
- Example 16: The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is not invertible because it is not bijective: $f(2) = f(-2) = 4$, so an inverse function cannot assign a unique value to 4. □

Composition of functions

- Let $g : A \to B'$ and $f : B \to C$ be functions such that the image of g is a subset of the domain of f. In particular, this holds if $B' \subseteq B$. The **composite function** of f with g, denoted by $f \circ g : A \to C$, is defined for every $a \in A$ by:

$$(f \circ g)(a) = f(g(a))$$

The function $f \circ g$ is called the **composition of f and g**.



Composition of functions

- Example 17: Let $f : \mathbb{Z} \to \mathbb{Z}$ and $g : \mathbb{Z} \to \mathbb{Z}$ be defined by

$$f(n) = n + 1 \qquad \text{and} \qquad g(n) = n^2.$$

Is it true that $f \circ g = g \circ f$? **Solution.** For every $n \in \mathbb{Z}$, we have

$$(f \circ g)(n) \;=\; f(g(n)) \;=\; f(n^2) \;=\; n^2 + 1.$$

However,

$$(g \circ f)(n) \;=\; g(f(n)) \;=\; g(n+1) \;=\; (n+1)^2 \;=\; n^2 + 2n + 1.$$

Therefore, $f \circ g \neq g \circ f$. □

- The example above shows that composition of functions is not commutative.

Composition with the identity function

- Given a domain A, the **identity function** $\iota_A : A \to A$ is defined by:

$$\iota_A(a) = a, \quad \forall a \in A.$$

- **Theorem.** If f is a function from X to Y, and ι_X and ι_Y are the identity functions on X and Y, respectively, then

$$\begin{aligned} f \circ \iota_X &= f \\ \iota_Y \circ f &= f \end{aligned}$$

- **Theorem.** If $f : X \to Y$ is a bijective function with inverse $f^{-1} : Y \to X$, then

$$\begin{aligned} f^{-1} \circ f &= \iota_X \\ f \circ f^{-1} &= \iota_Y \end{aligned}$$

Composition: Injectivity and surjectivity

- **Theorem.** If $f : X \to Y$ and $g : Y \to Z$ are injective functions, then $g \circ f$ is injective.

- **Theorem.** If $f : X \to Y$ and $g : Y \to Z$ are surjective functions, then $g \circ f$ is surjective.

Important functions: Floor and ceiling

- The **floor** function assigns to each real number x the greatest integer less than or equal to x. The value of the floor function at x is denoted by $\lfloor x \rfloor$.
- The **ceiling** function assigns to each real number x the least integer greater than or equal to x. The value of the ceiling function at x is denoted by $\lceil x \rceil$.
- Both functions have type $\mathbb{R} \to \mathbb{Z}$.
- Example 18: :
  - $\lfloor \pi \rfloor = 3$ $\quad \lceil \pi \rceil = 4$
  - $\lfloor -2.7 \rfloor = -3$ $\quad \lceil -2.7 \rceil = -2$
  - $\lfloor 42 \rfloor = 42$ $\quad \lceil 42 \rceil = 42$

□

Important functions: Floor and ceiling

- Some useful properties of the floor and ceiling functions:

  1. $\lfloor -x \rfloor = -\lceil x \rceil$
  2. $\lceil -x \rceil = -\lfloor x \rfloor$
  3. $x - 1 \quad < \quad \lfloor x \rfloor \quad \leq \quad x \quad \leq \quad \lceil x \rceil \quad < \quad x + 1$

- When proving properties of floor and ceiling functions, it is useful to observe that:

$$x = \lfloor x \rfloor + \varepsilon \qquad \text{for some } 0 \leq \varepsilon < 1,$$
$$x = \lceil x \rceil - \varepsilon \qquad \text{for some } 0 \leq \varepsilon < 1.$$

Important functions: Floor and ceiling

- Example 19: Let us prove that $\lfloor -x \rfloor = -\lceil x \rceil$. **Proof.** Write $x = n + \epsilon$, where $n \in \mathbb{Z}$ and $0 \leq \epsilon < 1$. There are two cases, depending on whether $\epsilon = 0$. Case 1: $\epsilon = 0$. In this case, $x = n$ and $\lfloor -x \rfloor = -\lceil x \rceil = -n$. Case 2: $0 < \epsilon < 1$. First, observe that

$$\begin{aligned} \lfloor -x \rfloor &= \lfloor -(n+\epsilon) \rfloor && \text{(since } x = n + \epsilon\text{)} \\ &= \lfloor -n - \epsilon \rfloor \\ &= -(n+1) && \text{(by the definition of floor),} \end{aligned}$$

and that

$$\begin{aligned} -\lceil x \rceil &= -\lceil n + \epsilon \rceil && \text{(since } x = n + \epsilon\text{)} \\ &= -(n+1) && \text{(by the definition of ceiling).} \end{aligned}$$

Hence, $\lfloor -x \rfloor = -\lceil x \rceil$. □

Partial functions

- A **partial function** f from a set A to a set B assigns a unique element b of B to each element a in a subset of A, called the **domain of definition of** f. The sets A and B are called the **domain** and **codomain** of f, respectively. We say that f is **undefined** at elements of A outside its domain of definition. When the domain of definition of f is the whole domain A, we say that f is a **total function**.

- Example 20: The function $f : \mathbb{Z} \to \mathbb{R}$ given by $f(n) = \sqrt{n}$ is a partial function from $\mathbb{Z}$ to $\mathbb{R}$ whose domain of definition is the set of non-negative integers. The function f is undefined at negative integers. □